

C2.5 What are the properties of the transition metals? (*separate science only*)

Teaching and learning narrative

The transition metals do not show group properties like the elements in Group 1 and Group 7; they form a family of elements with general properties that are different from those of other metals. These properties make the transition metals particularly useful. They all have relatively high melting points and densities.

Transitions metals are generally less reactive than Group 1 metals, and some are very unreactive (for example silver and gold).

Some transition metal elements and their compounds are used widely in the manufacture of consumer goods and as catalysts in industry, both of which represent beneficial applications of science (1aS4).

Assessable learning outcomes

Learners will be required to:

1. recall the general properties of transition metals (melting point, density, reactivity, formation of coloured ions with different charges and uses as catalysts) and exemplify these by reference to copper, iron, chromium, silver and gold

Linked learning opportunities

Practical work:

- investigate colours of transition metal compounds, and their effectiveness as catalysts